

清华大学本科生考试试题专用纸

考试课程： 大学物理（1）（英）

1. Choose the right answer for each question (3 pts each)

1.1 At 101325 Pa and 273.15 K, the total number of molecules for an ideal gas (diatomic molecules) within a volume of 1 m^3 is ($k_B = 1.38 \times 10^{-23}\text{ J/K}$):

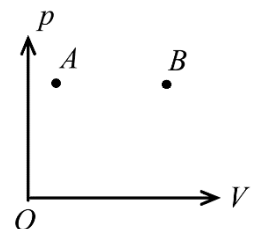
- (A) 6.02×10^{23} . (B) 6.02×10^{21} . (C) 2.69×10^{25} . (D) 2.69×10^{23} .

1.2 If $f(v)$ is the speed distribution of molecules, N is the total number of molecules, and m is the mass of each molecule, which of the following statement correctly describes the expression: $\int_{v_1}^{v_2} \frac{1}{2}mv^2 Nf(v)dv$?

- (A) It corresponds to the difference between the total translational kinetic energy of molecules with the speed v_2 and the total translational kinetic energy of molecules with the speed v_1 .
(B) It corresponds to the summation of the total translational kinetic energy of molecules with the speed v_2 and the total translational kinetic energy of molecules with the speed v_1 .
(C) It corresponds to the average translational kinetic energy of molecules with the speed in the range of v_1 to v_2 .
(D) It corresponds to the total translational kinetic energy of molecules with the speed in the range of v_1 to v_2 .

1.3 The ideal gas evolves from the equilibrium state A to another equilibrium state B, as shown on the right. Which of the following statement is correct, irrespective of the specific process taken for the gas?

- (A) The gas does positive work to the outside.
(B) The gas gains internal energy.
(C) The gas absorbs heat from the outside.
(D) The gas releases heat to the outside.



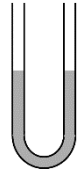
1.4 A horizontally stretched rope is fixed at one end and is attached to a motor at the other end. The motor oscillates up and down with a fixed frequency such that the rope carries a standing wave. By increasing the tensional force in the rope (we assume the extension of the rope stays unchanged), which of the following can happen?

- (A) The number of nodes of the standing wave increases.
(B) The number of nodes of the standing wave decreases.
(C) The number of nodes of the standing wave stays constant.
(D) The number of nodes of the standing wave increases first and then decreases.
(E) The number of nodes of the standing wave decreases first and then increases.

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1.5 A U-shaped tube with a constant cross-section is hanging vertically with its two openings facing upward. The liquid in the tube has a total length of L . What is the angular frequency of the liquid if it oscillates as a simple harmonic oscillator? (We neglect viscosity of the liquid and surface tension.)

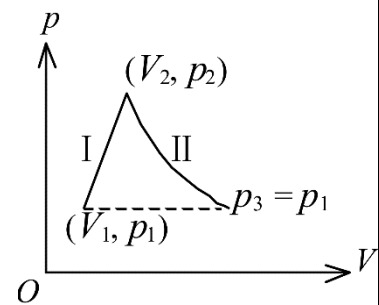


- (A) $\sqrt{g/L}$. (B) $\sqrt{2g/L}$. (C) $\sqrt{g/(2L)}$. (D) $2\sqrt{g/L}$.

2. Show the calculation steps and the final answers

2.1 (5 pts) A massless spring can extend by 30 cm under a pulling force of 60 N. The spring hangs vertically from the ceiling and its lower end is fixed to a block. We place a small object on the block such that the total weight of the block and the object is 4 kg. The block and the object are hanging still first. We now pull the block, together with the small object, downward from the balanced point by 10 cm. If we release the block from the pulled position and set it into oscillation, does the small object always stay on the block? (Let's assume that the block only moves up and down.)

2.2 (10 pts) Let us consider 1.0 mol of ideal gas of diatomic molecules. As shown on the right side, the system starts from a state with $p_1 = 1.01 \times 10^5$ Pa, $V_1 = 10^{-3} \text{ m}^3$ and takes the process I, a straight line in the $p-V$ plot, to the state with $p_2 = 4.04 \times 10^5$ Pa, $V_2 = 2 \times 10^{-3} \text{ m}^3$. After reaching (p_2, V_2) , it undergoes the process II following the equation: $pV^{1/2} = C$, where C is a constant, and reaches (p_3, V_3) where $p_3 = p_1$. Calculate the following quantities.



- (1) The heat absorbed by the gas in process I.
(2) The heat absorbed in the complete process (I + II).

2.3 (5 pts) In the Ruchardt experiment, a bead of mass m sits inside a narrow tube connected to a large vessel placed on the table. The vessel has a volume of V and the tube has a cross-sectional area s . The atmospheric pressure is p_0 and the gravitational acceleration is g . Initially, the bead is balanced inside tube due to the pressure difference between the air outside and the gas inside. Derive the oscillation frequency of the bead if it is perturbed from its balanced point. Suppose the bead can seal the gas inside the vessel but can move up and down without friction with the tube. Also, no heat exchange occurs for the gas ($\gamma = C_p/C_v$ for the gas inside the vessel).

